

Cloud Computing Practical

Examination

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Instructions

Perform all tasks using your Linux Virtual Machine.

Capture screenshots wherever required.

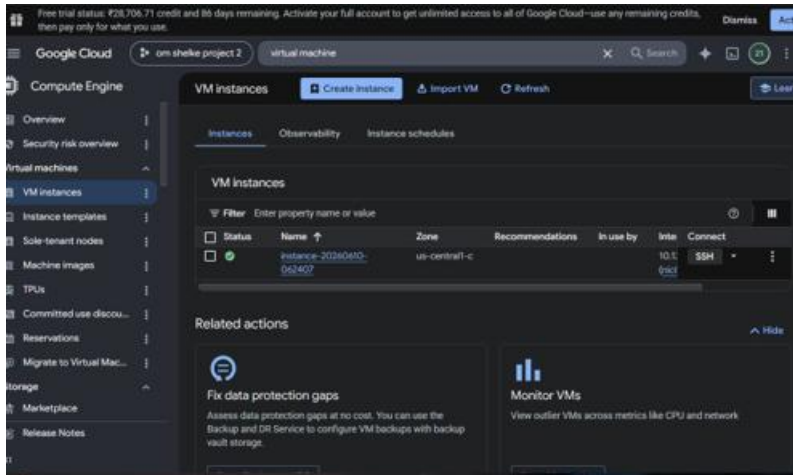
Store all shell scripts inside a folder named practical_exam.

Push all scripts and supporting files to a GitHub repository.

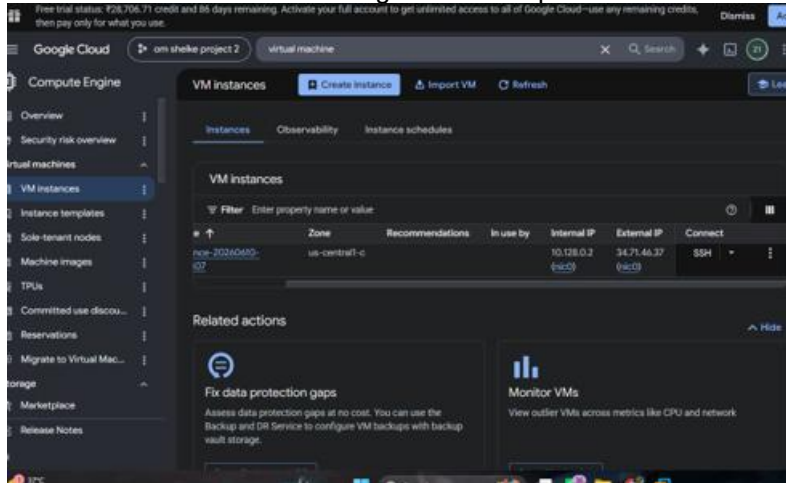
Submit a PDF report containing screenshots, outputs and GitHub repository URL.

Part A – Cloud & Linux VM Setup

Task 1: Create a Linux Virtual Machine with Public IP and SSH access enabled.



Task 2: Connect to the VM using SSH and capture screenshots.



Task 3: Create practical_exam/scripts, practical_exam/outputs and practical_exam/logs.

Part B – Linux Commands

Task 4: Display current working directory, current user and hostname. Save output.

mkdir

```
Linux instance-20260610-080345 6.1.0-49-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.174-1 (2026-05-26) x86_64
```

```
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.
```

```
guttasantosh44@instance-20260610-080345:~$ mkdir practical_exam
```

```
guttasantosh44@instance-20260610-080345:~$ pwd  
/home/guttasantosh44
```

```
guttasantosh44@instance-20260610-080345:~$
```

Task 5: Create 5 files inside scripts folder.



SSH-in-browser



UPLOAD FILE



DOWNLOAD FILE



```
guttasantosh44@instance-20260610-062407:~$ cd practical_exam
guttasantosh44@instance-20260610-062407:~/practical_exam$ mkdir scripts
guttasantosh44@instance-20260610-062407:~/practical_exam$ cd...
-bash: cd...: command not found
guttasantosh44@instance-20260610-062407:~/practical_exam$ cd ..
guttasantosh44@instance-20260610-062407:~$ mkdir practical_output
guttasantosh44@instance-20260610-062407:~$ cd practical_exam
guttasantosh44@instance-20260610-062407:~/practical_exam$ dir
scripts
guttasantosh44@instance-20260610-062407:~/practical_exam$ cd scripts
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts$ touch
one.txt two.txt three.txt four.txt five.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts$ ls
five.txt four.txt one.txt three.txt two.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts$
```

Task 6: Copy two files into outputs folder.

Task 7: Rename one file.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ mv one
txt six.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ls
six.txt two.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 8: Delete one file.

```
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ rm six.txt
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ls
two.txt
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 9: Display hidden files.

```
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ls -a
.  ..  two.txt
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 10: Display disk usage information.

```
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ls -a
.  ..  two.txt
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M  536K  392M   1% /run
/dev/sda1       9.7G  2.7G  6.6G  29% /
tmpfs           2.0G   0    2.0G   0% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
/dev/sda15      124M  12M  113M  10% /boot/efi
tmpfs           393M   0    393M   0% /run/user/1000
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 11: Display memory usage information.

```
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.0Gi          532Mi        2.4Gi        536Ki        1.1Gi
Swap:          0B           0B           0B
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 12: Display running processes.

```
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ps
  PID TTY          TIME CMD
 1597 pts/0    00:00:00 bash
 3233 pts/0    00:00:00 ps
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.1  0.3 167976 12544 ?        Ss   06:33   0:02 /sbin/i
root         2   0.0  0.0      0     0 ?        S    06:33   0:00 [kthrea
root         3   0.0  0.0      0     0 ?        I<   06:33   0:00 [rcu_gp
gutesantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 13: Find all .sh files inside practical_exam.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ find . -type f -name "*.sh"
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Part C – Permissions & Users

Task 14: Create secure_data.txt and apply permission 640.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ touch secure_data.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ chmod 640 secure_data.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 15: Demonstrate Owner, Group and Others using commands.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ ls -l secure_data.txt
-rw-r----- 1 guttasantosh44 guttasantosh44 0 Jun 10 07:02 secure_data.txt
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 16: Display User ID, Group ID and assigned groups.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ id
uid=1000(guttasantosh44) gid=1003(guttasantosh44) groups=1003(guttasantosh44),4(adm),30(dip),44(video),46(plugdev),1000(google-sudoers),1001(docker),
1002(lxd)
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Part D – Shell Scripting

Task 17: Script to print Welcome to Cloud Computing.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ echo "Welcome to Cloud Computing"
Welcome to Cloud Computing
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 18: Script accepting username and displaying Welcome .

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ read -p "Enter your username: " username
echo "Welcome, $username!"
Enter your username: om
Welcome, om!
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 19: Script to find larger of two numbers using if-else.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ read -p "Enter first number: " num1
read -p "Enter second number: " num2
Enter first number: 10
Enter second number: 20
20 is larger than 10.
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 20: Script to determine even or odd number.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$ read -p "Enter a number: " num
if [ $((num % 2)) -eq 0 ]; then
    echo "$num is Even."
else
    echo "$num is Odd."
fi
Enter a number: 10
10 is Even.
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 21: For loop displaying 1 to 20.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$  
#!/bin/bash  
for i in {1..20}  
do  
    echo $i  
done  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$
```

Task 22: While loop displaying 20 to 1.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$  
#!/bin/bash  
count=20  
while [ $count -ge 1 ]  
do  
    echo $count  
    ((count--))  
done  
20  
19  
18  
17  
16  
15  
14  
13  
12  
11  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$
```


Task 23: Multiplication table script.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$  
#!/bin/bash  
read -p "Enter a number for the multiplication table: " num  
  
for i in {1..10}  
do  
    echo "$num x $i = $((num * i))"  
done  
Enter a number for the multiplication table: 50  
50 x 1 = 50  
50 x 2 = 100  
50 x 3 = 150  
50 x 4 = 200  
50 x 5 = 250  
50 x 6 = 300  
50 x 7 = 350  
50 x 8 = 400  
50 x 9 = 450  
50 x 10 = 500  
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/outputs$
```

Task 24: Count files inside scripts directory.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
# Lists all filenames ending in .sh in the current directory  
echo "Shell script files in the current directory:"  
ls *.sh 2>/dev/null || echo "No .sh files found."  
Shell script files in the current directory:  
█
```

Task 25: Display all shell script filenames.

Task 26: Display current date and time and save output.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$  
date  
Wed Jun 10 07:13:42 UTC 2026  
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Part E – Processes & Scheduling

Task 27: Save running processes to outputs/processes.txt.

```
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$  
# Ensure the outputs directory exists  
mkdir -p outputs  
  
# Save the running processes  
ps aux > outputs/processes.txt  
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 28: Identify top 5 memory-consuming processes.


```

guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$
ps aux --sort=-%mem | head -n 6
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          1442  1.3  3.9 2073180 158912 ?        Ssl   06:33    0:34 /opt/go
ogle-cloud-ops-agent/subagents/opentelemetry-collector/otelopscol --config
=/run/google-cloud-ops-agent-opentelemetry-collector/otel.yaml
root          1440  0.0  1.5 1868812 64188 ?        Ssl   06:33    0:00 /opt/go
ogle-cloud-ops-agent/libexec/google_cloud_ops_agent_wrapper --config_path /
etc/google-cloud-ops-agent/config.yaml --log_path /var/log/google-cloud-ops
-agent/subagents/logging-module.log /opt/google-cloud-ops-agent/subagents/
fluent-bit/bin/fluent-bit --config /run/google-cloud-ops-agent-fluent-bit/
fluent_bit_main.conf --parser /run/google-cloud-ops-agent-fluent-bit/fluen
t_bit_parser.conf --storage_path /var/lib/google-cloud-ops-agent/fluent-bi
t/buffers
root           388  0.0  1.5 1288792 60916 ?        Ssl   06:33    0:01 /usr/bi
n/google_osconfig_agent
root           1134  0.0  0.6 1279572 27928 ?        Sl    06:33    0:01 /usr/li
b/google/guest_agent/GuestAgentCorePlugin/core_plugin --protocol=unix --ad
dress=/run/google-guest-agent/plugin-connections/GuestAgentCorePlugin_2026
0511.00.sock --errorlogfile=/var/lib/google-guest-agent/481642438515207268
7/agent_state/plugins/GuestAgentCorePlugin/plugin.log
root           1013  0.0  0.6 1278188 26924 ?        Ssl   06:33    0:02 /usr/bi
n/google_guest_agent_manager
guttasantosh44@instance-20260610-062407:~/practical_exam/scripts/output$

```

Task 29: Create a cron job appending current date and time every minute.

```
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

Task 30: Verify cron execution with evidence.

```
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
~
~
~
~
~
~
~
~
~
~
~
~
~
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~
~
~
~
~
~
~
```

Part F – Git & Submission

Task 31: Create GitHub repository cloud-practical-exam.

Task 32: Initialize Git in practical_exam directory.

Task 33: Add all files.

Task 34: Commit changes with message Cloud Practical Exam Submission.

Task 35: Push files to GitHub.

Task 36: Verify repository contains scripts, outputs, screenshots and documentation.

Final Deliverables

PDF report containing screenshots and outputs.

GitHub Repository URL.

All shell scripts.

Evidence of cron execution.

Evidence of successful SSH connection.